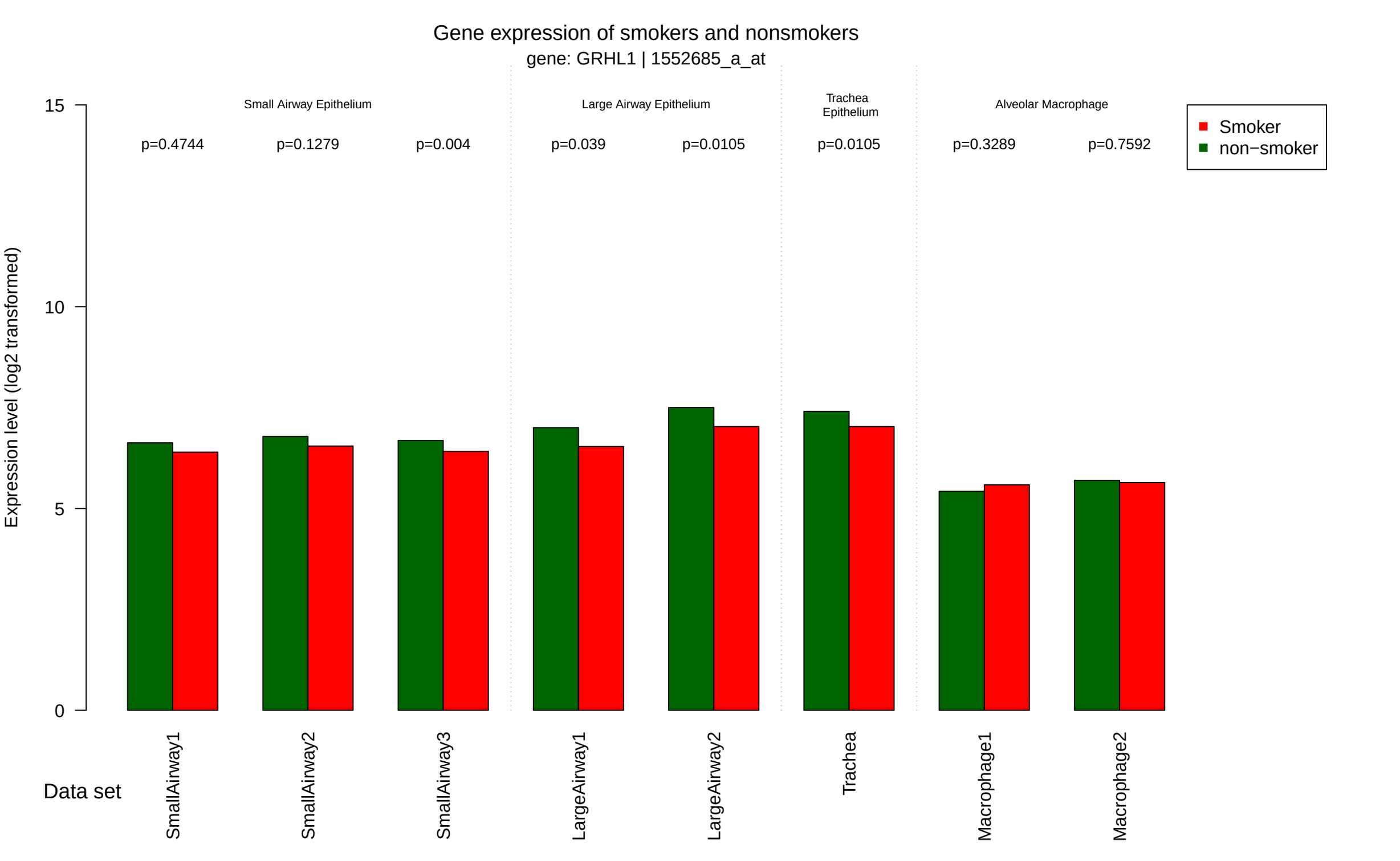
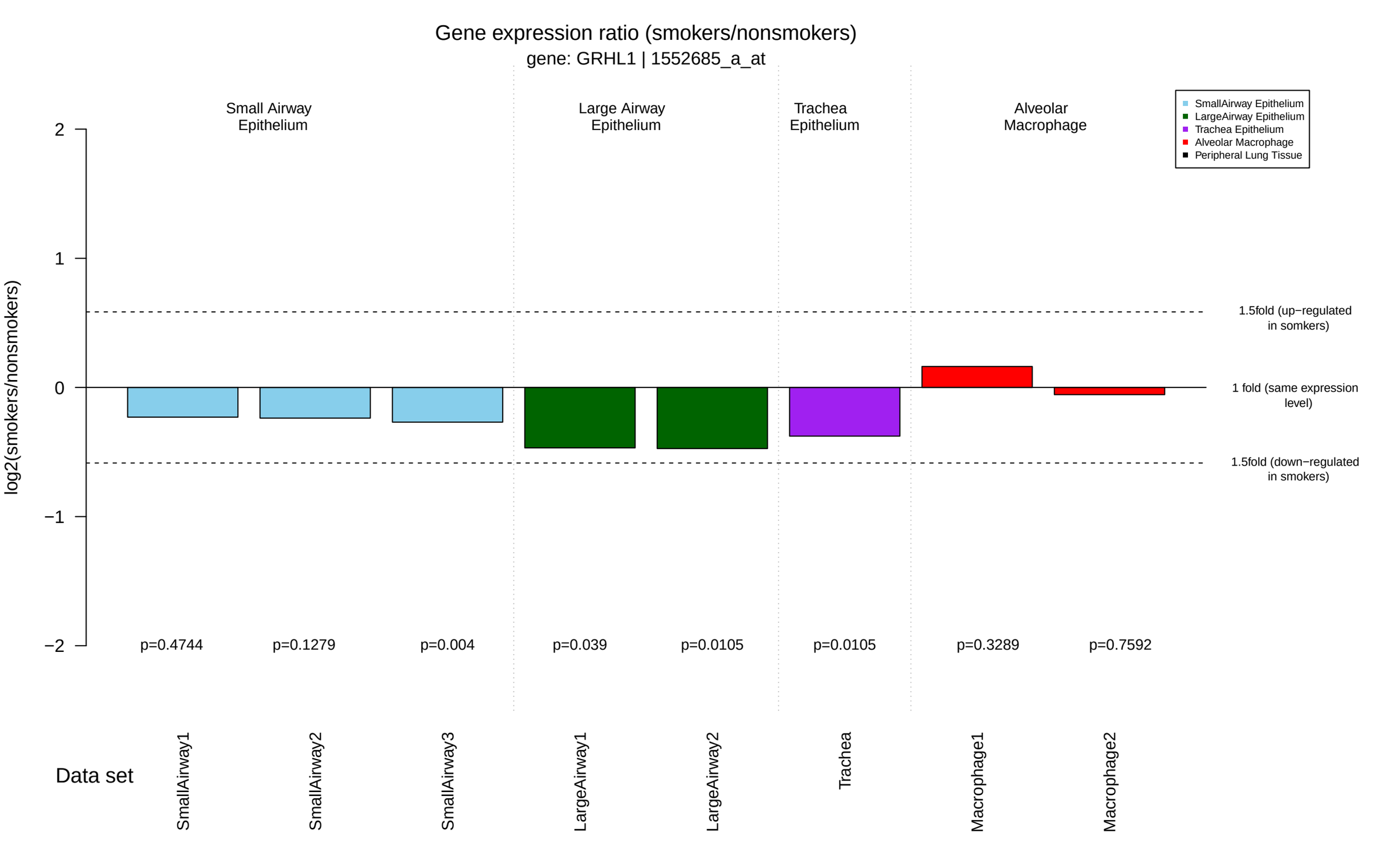
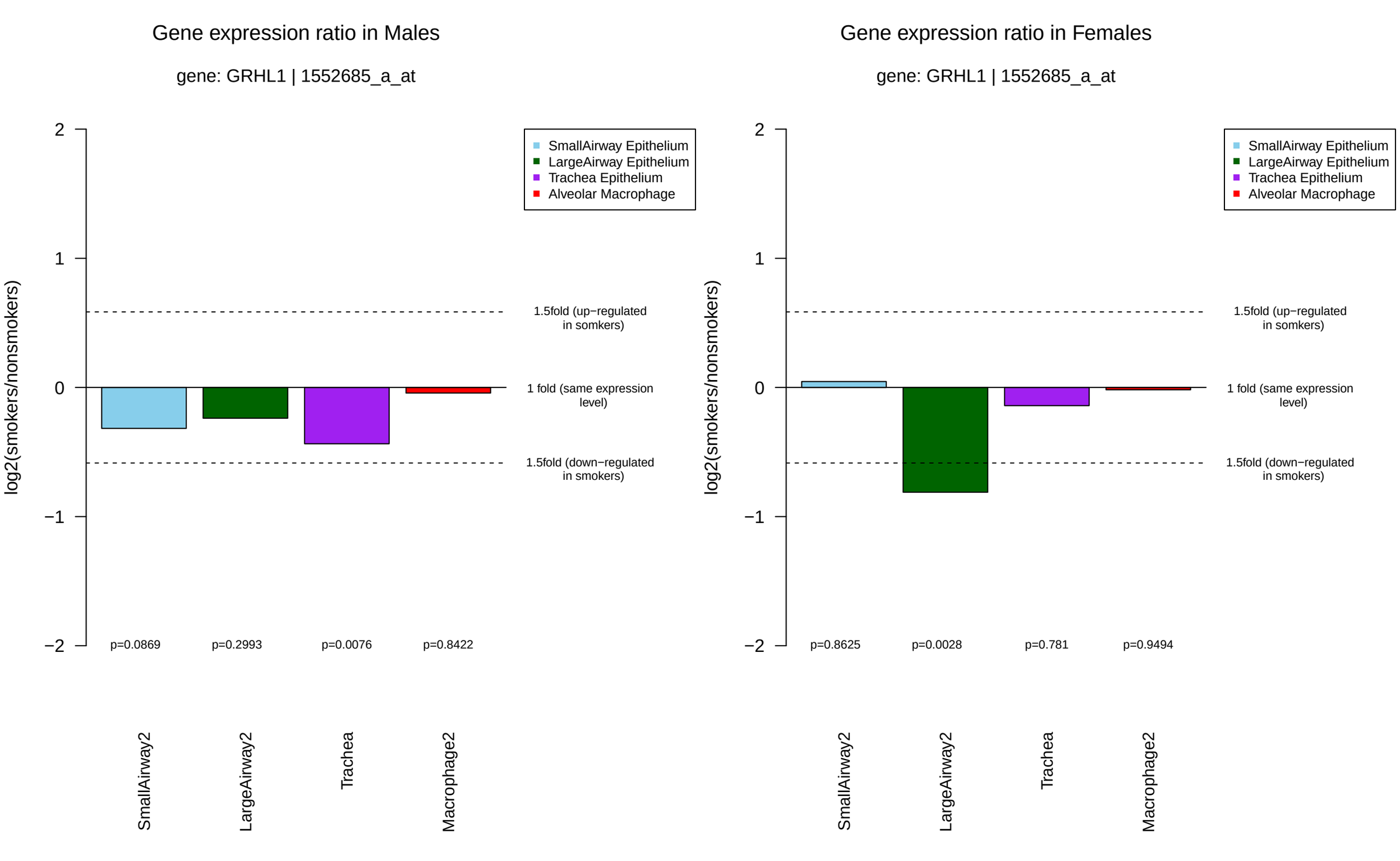
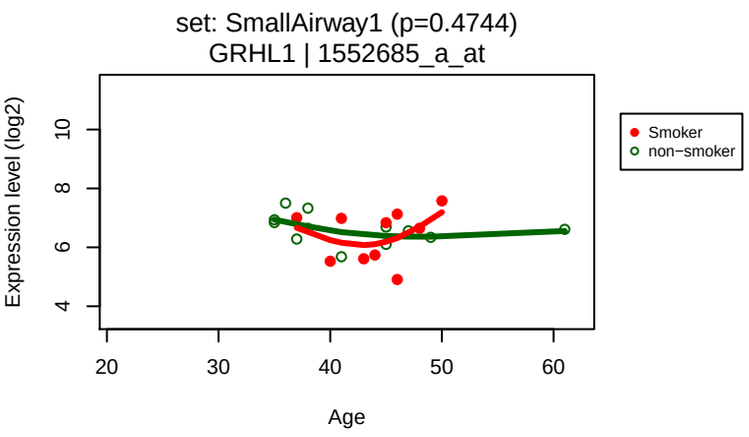
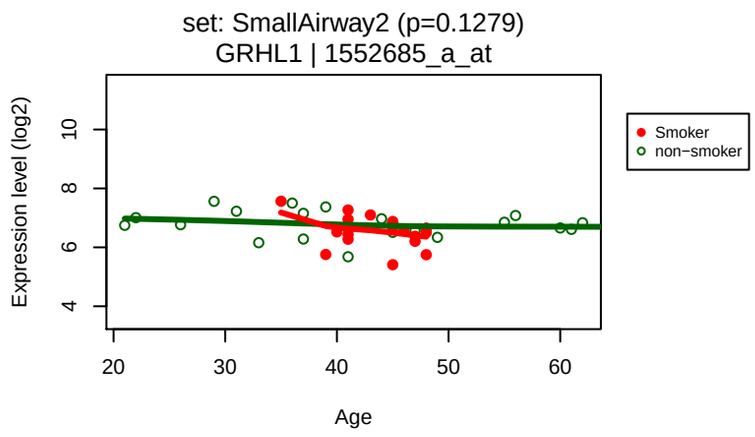
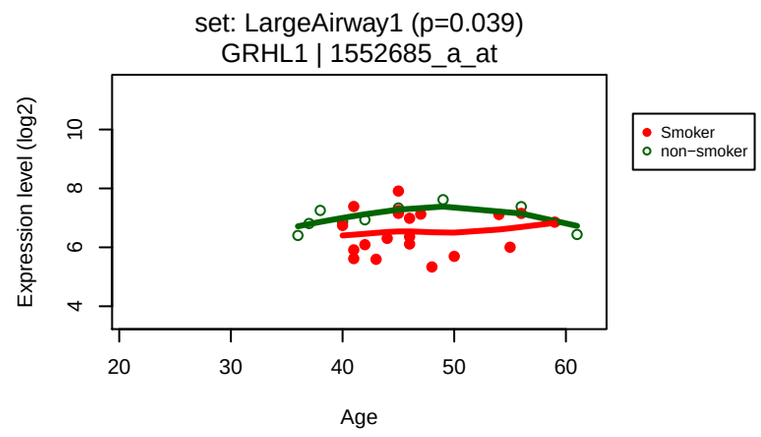

Gene expression of smokers and nonsmokers
gene: GRHL1 | 1552685_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4744
p=0.1279
p=0.004
p=0.039
p=0.0105
p=0.0105
p=0.3289
p=0.7592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

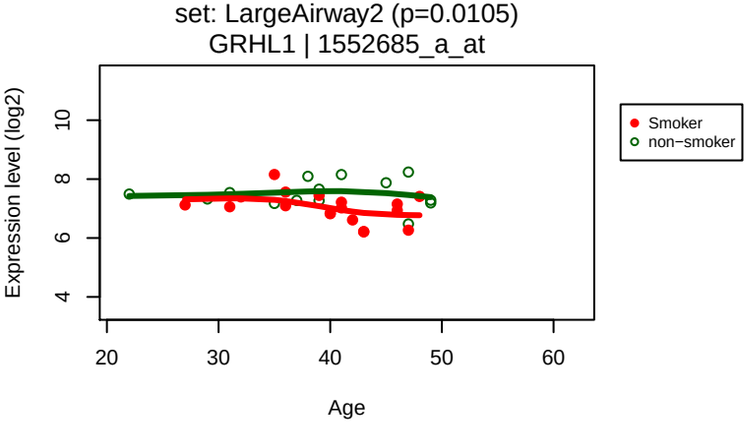

Gene expression ratio (smokers/nonsmokers)
gene: GRHL1 | 1552685_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4744
p=0.1279
p=0.004
p=0.039
p=0.0105
p=0.0105
p=0.3289
p=0.7592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

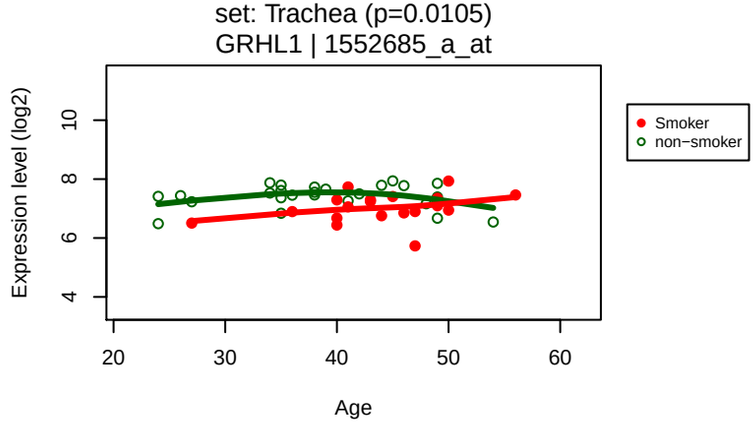

Gene expression ratio in Males
gene: GRHL1 | 1552685_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0869
p=0.2993
p=0.0076
p=0.8422
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRHL1 | 1552685_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8625
p=0.0028
p=0.781
p=0.9494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

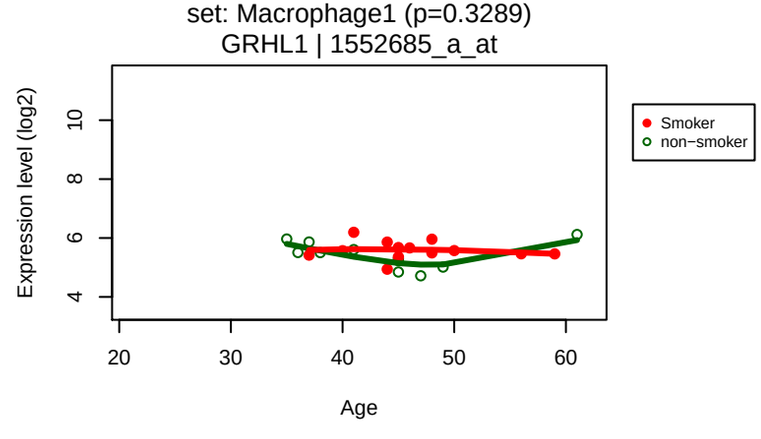

set: SmallAirway1 (p=0.4744)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker

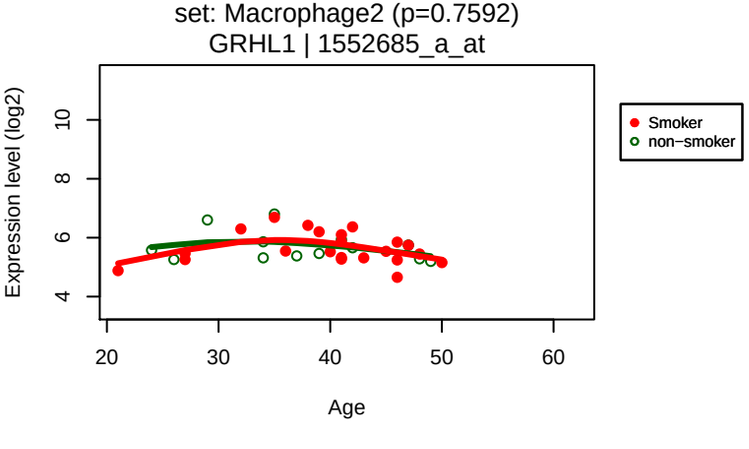

set: SmallAirway2 (p=0.1279)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker

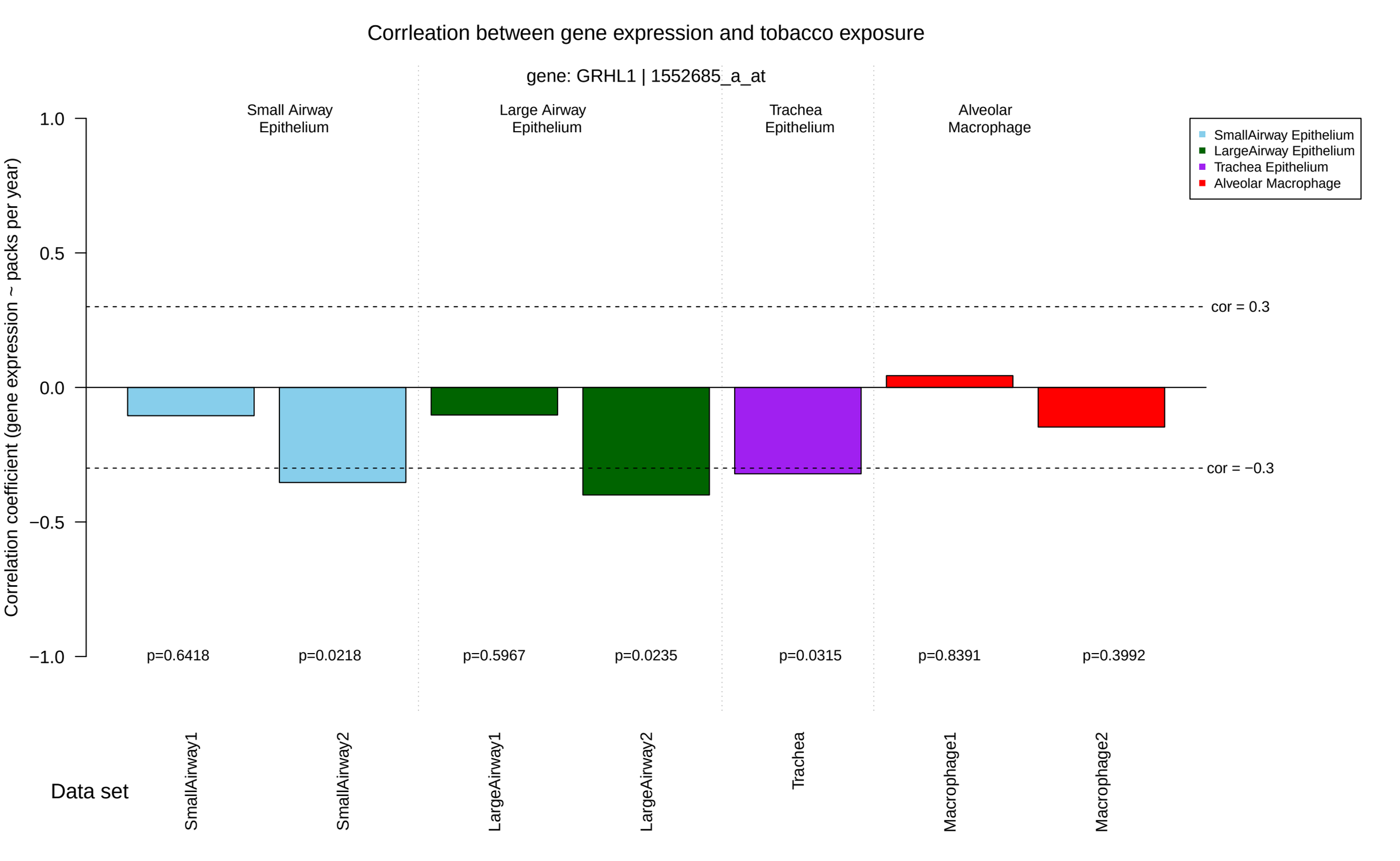

set: LargeAirway1 (p=0.039)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0105)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0105)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3289)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7592)
GRHL1 | 1552685_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRHL1 | 1552685_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6418
p=0.0218
p=0.5967
p=0.0235
p=0.0315
p=0.8391
p=0.3992
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2